25901234 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEECT205 - CONTINUOUS AND DISCRETE SYSTEMS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **10** questions

Each question carries **2** marks

1. Write a short note on the IC 741 op-amp. [K] / [CO1]
2. Which BJT configuration is most popular for amplification and why? [K] / [CO1]
3. What is an Op-amp? Draw the schematic symbol of an op-amp. [K] / [CO1]
4. State the Barkhausen criteria. [K] / [CO1]
5. Which component mainly decide the time constant in an op amp integrator [U] / [CO2]
6. Which component from the frequency determining network in an op amp colpitts oscillator [U] / [CO2]
7. What is meant by a subtractor circuit using an op-amp? [U] / [CO2]
8. What is a counter? [K] / [CO3]
9. Find 1’s compliment of the number 10011011 [U] / [CO3]
10. Subtract the binary numbers: 1011 − 0101 [U] / [CO3]

11. Simplify the expression: Y = A + AB. [A] / [CO3]

12. Describe the importance of logic gates in digital electronics. [K] / [CO3]

13. What is the difference between Binary and Hexadecimal number system? [K] / [CO3]

14. What are minterms? [K] / [CO3]

[2x10 = 20]

Part B

Short Essay

Answer any **6** questions

Each question carries **5** marks

15. Explain RC coupled amplifier with circuit diagram. [U] / [CO1]

16. If Rf = 100 k ohm and R1 = 10k ohm, calculate the gain for both inverting and non- inverting configurations of op-amp. [U] / [CO1]

17. Define (a) Input Offset Voltage (b) Input Offset Current (c) CMRR (d) Slew Rate [K] / [CO1]

18. List important features of the 555 Timer and draw the pin diagram. [U] / [CO2]

19. Design a High pass filter for the cut off frequency of 2 KHz and draw its circuit diagram and frequency response [U] / [CO2]

20. Compare SR and JK flip-flops. [U] / [CO3]

21. Describe the role of truth tables in understanding logic gate operations. [U] / [CO3]

22. What is the equivalent expression for the Boolean expression x'y'z +yz+ xz? [A] / [CO3]

[5x6 = 30]